Badminton Training System - User Survey

Research Study Information

Study Title: Evaluation of a Real-Time Badminton Training System for Shot Tracking and Performance Analytics

Purpose: This survey aims to evaluate the usability, perceived usefulness, and user satisfaction of the Badminton Training System. Your responses will help identify strengths and areas for improvement.

Estimated Time: 10-15 minutes

Confidentiality: All responses are anonymous and will be used solely for research purposes.

Informed Consent

By proceeding with this survey, you acknowledge that:

- ☐ I am 18 years of age or older
- ☐ I understand the purpose of this study
- ☐ My participation is voluntary and I may withdraw at any time
- ☐ My responses will be kept confidential and anonymous
- ☐ I consent to participate in this research study

Section 1: System Usability Scale (SUS)

The System Usability Scale is a validated 10-item questionnaire (Brooke, 1996). Please indicate your level of agreement with each statement.

Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

#	Statement	1	2	3	4	5
1	I think that I would like to use this system frequently					
2	I found the system unnecessarily complex					
3	I thought the system was easy to use					
4	I think that I would need the support of a technical person to be able to use this system					
5	I found the various functions in this system were well integrated					
6	I thought there was too much inconsistency in this system					
7	I would imagine that most people would learn to use this system very quickly					
8	I found the system very cumbersome to use					

#	Statement	1	2	3	4	5
9	I felt very confident using the system					
10	I needed to learn a lot of things before I could get going with this system					

> **Reference:** Brooke, J. (1996). SUS: A "Quick and Dirty" Usability Scale. In P. W. Jordan et al. (Eds.), *Usability Evaluation in Industry* (pp. 189-194). Taylor & Francis.

Section 2: Technology Acceptance Model (TAM)

The following questions are adapted from the Technology Acceptance Model (Davis, 1989). Please indicate your level of agreement with each statement.

Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

Perceived Usefulness (PU)

#	Statement	1	2	3	4	5
1	Using the system improves my training performance					
2	Using the system increases my training productivity					
3	Using the system enhances my effectiveness in training					
4	Using the system makes it easier to track training progress					
5	I find the system useful for badminton training					
6	The system provides valuable insights into shot accuracy					

Perceived Ease of Use (PEOU)

#	Statement	1	2	3	4	5
7	Learning to operate the system was easy for me					
8	I find it easy to get the system to do what I want it to do					
9	My interaction with the system is clear and understandable					
10	I find the system flexible to interact with					
11	It was easy for me to become skillful at using the system					
12	I find the system easy to use					

Behavioral Intention to Use (BI)

#	Statement	1	2	3	4	5
13	I intend to continue using the system in the future					

#	Statement	1	2	3	4	5
14	I would recommend the system to other coaches/athletes					
15	I plan to use the system regularly for training sessions					

> **Reference:** Davis, F. D. (1989). Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. *MIS Quarterly*, 13(3), 319-340.

Section 3: Feature-Specific Evaluation

Please rate your satisfaction with each feature of the system.

Scale: 1 = Very Dissatisfied, 2 = Dissatisfied, 3 = Neutral, 4 = Satisfied, 5 = Very Satisfied, N/A = Did not use

#	Feature	1	2	3	4	5	N/A
1	User authentication (login/register)						
2	Athlete profile management						
3	Training session controls (start/stop)						
4	Real-time shot tracking						
5	Target template selection						
6	Court visualization display						
7	Shot accuracy indicators						
8	Performance statistics dashboard						
9	Session history and records						
10	Session detail view						

Section 4: System Performance

Please rate your agreement with the following statements about system performance.

Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

#	Statement	1	2	3	4	5
1	The system responds quickly to my actions					
2	Shot data updates in real-time without noticeable delay					
3	The system is reliable and rarely experiences errors					
4	My training data is accurately recorded and saved					
5	The system performs consistently across sessions					

Section 5: Training Impact

Please rate your agreement with the following statements about the system's impact on training.

Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

#	Statement	1	2	3	4	5
1	The system helps me identify areas for improvement					
2	The data provided helps me make better training decisions					
3	Using the system has increased my motivation to train					
4	The system has improved communication between coach and athlete					
5	I can track my progress over time using the system					
6	The accuracy feedback helps me adjust my technique					

Section 6: Net Promoter Score (NPS)

On a scale of 0-10, how likely are you to recommend the Badminton Training System to a colleague or friend?

0	1	2	3	4	5	6	7	8	9	10
Not at all likely					Neutral					Extremely likely

> **Scoring:** 0-6 = Detractor, 7-8 = Passive, 9-10 = Promoter NPS = % Promoters - % Detractors

Section 7: Future Features (Priority Ranking)

Please rate the importance of the following potential features.

Scale: 1 = Not Important, 2 = Slightly Important, 3 = Moderately Important, 4 = Important, 5 = Very Important

#	Potential Feature	1	2	3	4	5
1	Video recording integration with shot data					
2	Shot heatmap visualization					
3	Session replay/animation feature					
4	Performance comparison between athletes					
5	Goal setting and progress tracking					
6	AI-powered training recommendations					
7	Export reports (PDF/CSV)					

#	Potential Feature	1	2	3	4	5
8	Mobile application					
9	Integration with wearable devices					
10	Social features (leaderboards, sharing)					

Section 8: Open-Ended Questions

Please provide your detailed feedback on the following questions.

8.1 What aspects of the system do you find most valuable for your training?

8.2 What challenges or difficulties have you experienced while using the system?

8.3 What improvements would you suggest for the system?

8.4 Do you have any additional comments or feedback?

Section 9: Demographics

The following questions help us understand our user base. All responses are optional and anonymous.

9.1 What is your primary role?

- ☐ Coach
- ☐ Athlete
- ☐ Both Coach and Athlete
- ☐ Other: _____
- ☐ Prefer not to say

9.2 What is your age range?

- ☐ 18-24
- ☐ 25-34
- ☐ 35-44
- ☐ 45-54
- ☐ 55-64
- ☐ 65 or older
- ☐ Prefer not to say

9.3 What is your gender?

- ☐ Male
- ☐ Female
- ☐ Non-binary
- ☐ Other: _____
- ☐ Prefer not to say

9.4 How many years of badminton experience do you have?

- ☐ Less than 1 year
- ☐ 1-3 years
- ☐ 4-6 years
- ☐ 7-10 years
- ☐ More than 10 years
- ☐ Prefer not to say

9.5 What is your current skill level?

- ☐ Beginner
- ☐ Intermediate
- ☐ Advanced
- ☐ Professional/Competitive
- ☐ Prefer not to say

9.6 How frequently do you train per week?

- ☐ 1-2 times
- ☐ 3-4 times
- ☐ 5-6 times
- ☐ Daily
- ☐ Prefer not to say

9.7 How long have you been using the Badminton Training System?

- ☐ Less than 1 week
- ☐ 1-4 weeks
- ☐ 1-3 months
- ☐ More than 3 months
- ☐ This is my first time using it

9.8 Have you used other digital training tools before?

- ☐ Yes
- ☐ No
- ☐ Prefer not to say

Section 10: Follow-Up (Optional)

10.1 Would you be willing to participate in a follow-up interview for more detailed feedback?

- ☐ Yes
- ☐ No

10.2 If yes, please provide your email address:

Your email will only be used to contact you for the follow-up interview and will be stored separately from your survey responses to maintain anonymity.

End of Survey

Thank you for completing this survey!

Your feedback is invaluable in improving the Badminton Training System. If you have any questions about this research, please contact: [Researcher Contact Information]

Survey Scoring Guide (For Researcher)

SUS Score Calculation

1. For items 1, 3, 5, 7, 9: Score = Response - 1
2. For items 2, 4, 6, 8, 10: Score = 5 - Response
3. Sum all scores and multiply by 2.5
4. Final score range: 0-100 (>68 = above average usability)

TAM Score Calculation

- **Perceived Usefulness (PU):** Average of items 1-6
- **Perceived Ease of Use (PEOU):** Average of items 7-12
- **Behavioral Intention (BI):** Average of items 13-15
- Higher scores indicate stronger agreement (range: 1-5)

NPS Calculation

- NPS = (% of Promoters) - (% of Detractors)
- Score range: -100 to +100
- Positive score = Good, >50 = Excellent

Reliability Testing

- Calculate Cronbach's Alpha for each scale
- Acceptable threshold: $\alpha \geq 0.70$

References

1. Brooke, J. (1996). SUS: A "Quick and Dirty" Usability Scale. In P. W. Jordan, B. Thomas, B. A. Weerdmeester, & I. L. McClelland (Eds.), *Usability Evaluation in Industry* (pp. 189-194). Taylor & Francis.

2. Davis, F. D. (1989). Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. *MIS Quarterly*, 13(3), 319-340. https://doi.org/10.2307/249008

3. Reichheld, F. F. (2003). The One Number You Need to Grow. *Harvard Business Review*, 81(12), 46-54.

4. Bangor, A., Kortum, P. T., & Miller, J. T. (2008). An Empirical Evaluation of the System Usability Scale. *International Journal of Human-Computer Interaction*, 24(6), 574-594.